

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

sdkman chrushed my ide

hibernate task not forkin

so
all **good** no used to
far! good well yet
no problems anymore

Today's agenda

- Total Recall
- Hibernate
- Quiz
- Questions

Total Recall



What do you remember from the last session?

factory jpa i use vscode
entity manager

What is 'Database normalization'?

3nf reduce redundancy
references
improve data integrity
efn
organizing data
atomic transactions
reducing data redundancy
3f third normal form

What is isolation in frames of databases?

reliable transaction proc
serializable **acid** transaction
i in acid salamaleykum
independence

What is the difference between fetching Lazy vs Eager ?

loads required
on-demand

lazy loads on demand
immediately or on demand

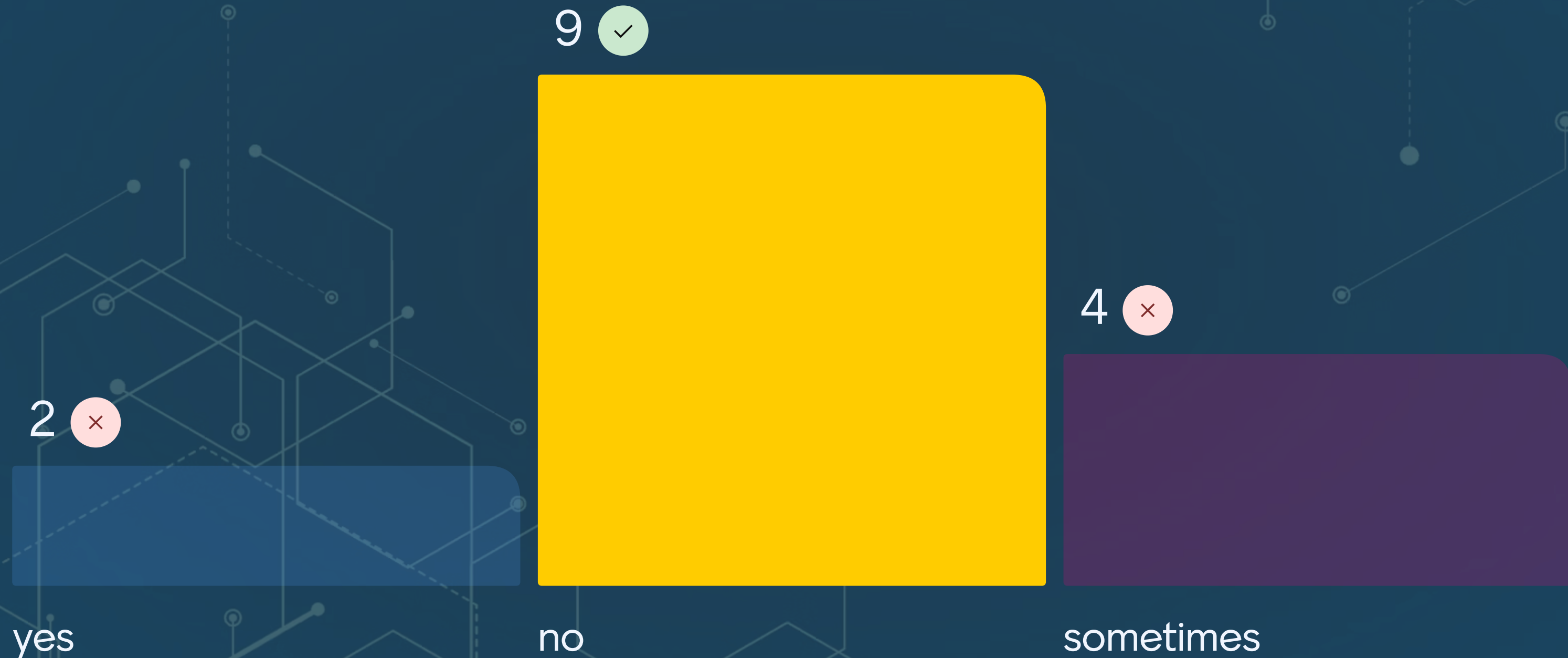
eager loads related data
immediately

loads all

load data now or later

vs

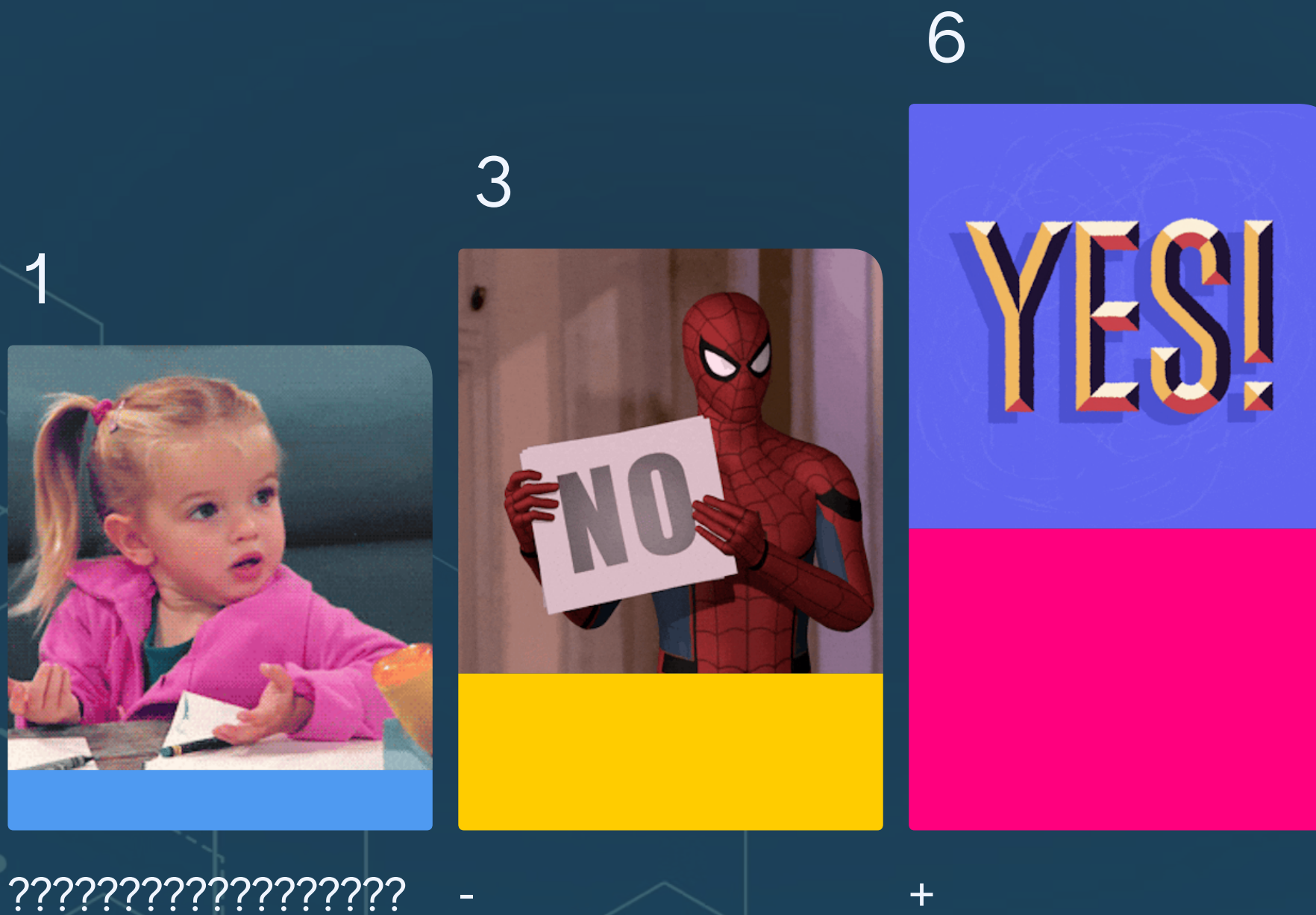
Is it correct to initialize services in each other?



Hibernate (ws-07)

Normalization | Isolation | Transaction | Query

Have you started the theory of 'Hibernate'?



Explore module ws-07

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026>

10 mins beak

Quiz

What does JPA stand for?

17 ✓



5 ✓



0 ✗

1 ✗



Java Programming API

Java Persistence API

Jakarta Persistence API

entityManager.start();

What is the basic difference between JPA and Hibernate

24 ✓



1 ✗

JPA is native

0 ✗

JPA is a specification; Hibernate is an implementation

Hibernate is for professional use

What is TRUE about @Transactional annotation in JPA?

21 ✓

4 ✗

It ensures that all operations within method are committed to database

0 ✗

It auto creates new instances of entities in persistence context

0 ✗

Method will run in scope of a transaction and commit or rollback

Enforces to flush changes to database every time a query is executed

Differences Between JPA and Hibernate ?

JPA is a specification;
Hibernate is an
implementation

JPA is a specification;
Hibernate is an
implementation of that
specification.

JPA is a specification;
Hibernate is an
implementation of JPA.

JPA is a specification
defining the standard rules
for ORM, while Hibernate is
a concrete framework

JPA - factory while
Hibernate Session

JPA is a specification;
Hibernate is an
implementation of JPA.

Portability and
standardized

hibernate is an
implementation and more
efficient, jpa more flexible

Differences Between JPA and Hibernate ?

JPA = is a specification,
Hibernate = is an
implementation of JPA

jpa is specs, hibernate
tech

JPA is a specification,
Hibernet is an
implementation

first is specification and
2nd is an
implementation of it

JPA is the abstract standard
and interface defining how
ORM should work in Java, while
Hibernate is the actual
concrete engine that
implements that standard

JPA uses EntityManager
and EntityManagerFactory
Hibernate uses Session and
SessionFactory

JPA is the specification,
Hibernate is the
implementation.

JPA defines how Java objects
are persisted while Hibernate is
an implementation of JPA that
actually performs the ORM and
DB operations

Differences Between JPA and Hibernate ?

JPA = rules/interface
Hibernate = implementation
that follows those rules

JPA has basic DDL.
Hibernate offers
advanced auto-
generation

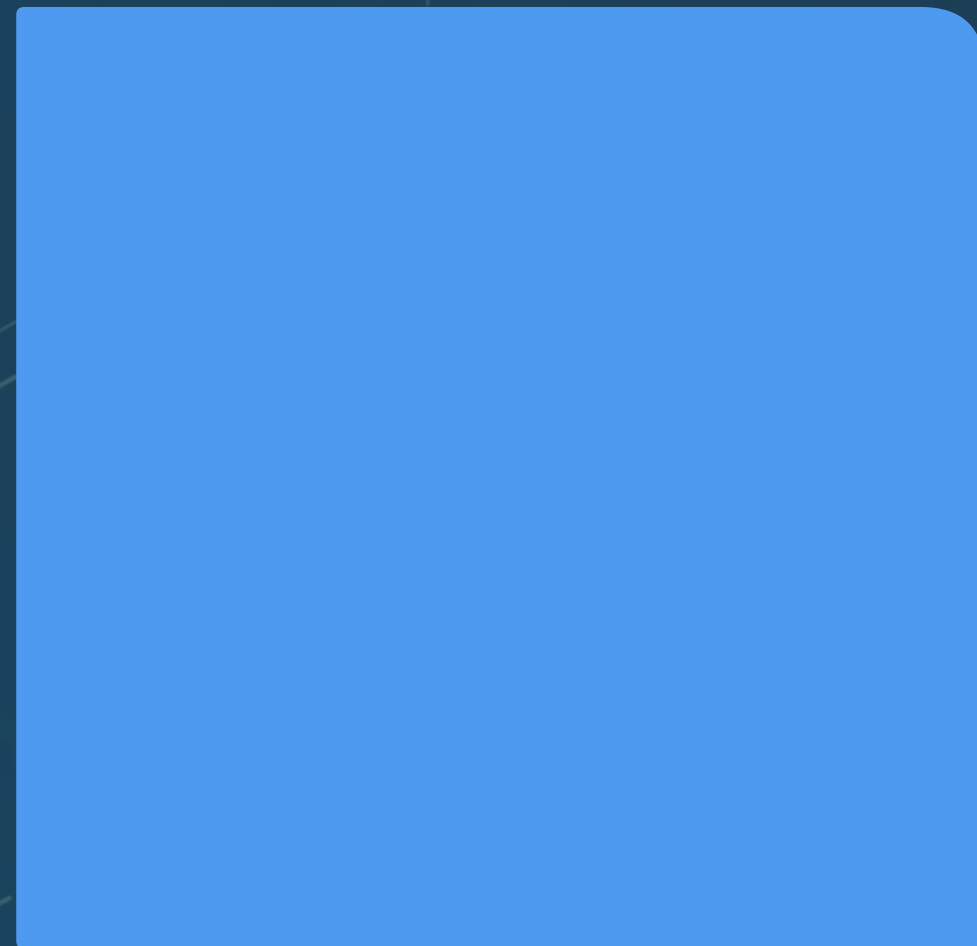
JPA is a specification for ORM
in Java, while Hibernate is a
concrete ORM framework and
implementation of the JPA
specification that provides the
actual working code.

Differences Between JPA and Hibernate

1. JPA is a specification while Hibernate is an actual implementation of the JPA specification.
2. Hibernate implements all JPA features and adds its own enhancements.
3. Choosing native Hibernate leads to less portability but better performance.

Can we use EntityManager, while leveraging Hibernate?

23 ✓



yes

2 ✗

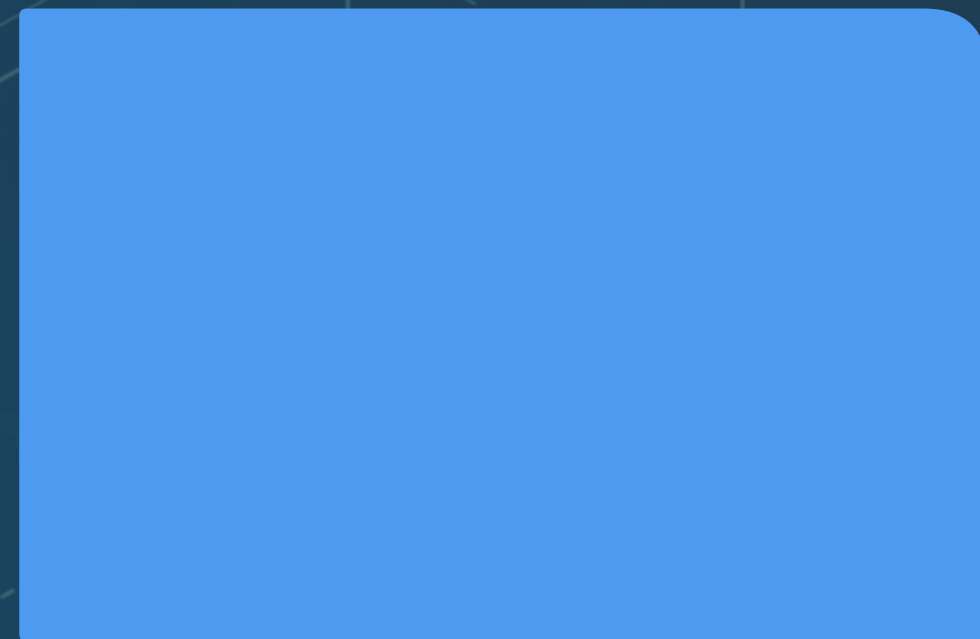
no

1 ✗

in some cases

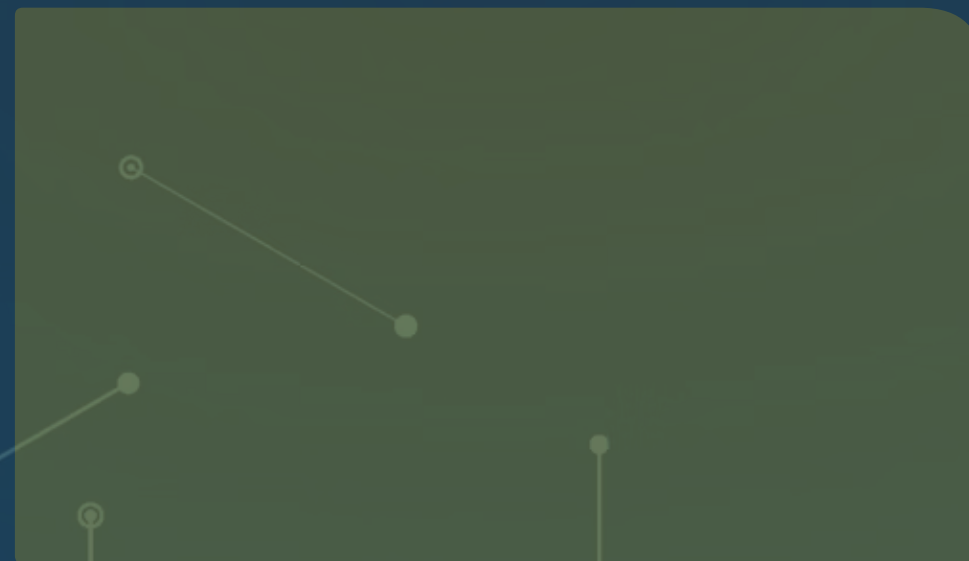
Can we say that EntityManager is the same as Session, when leveraging Hibernate?

8



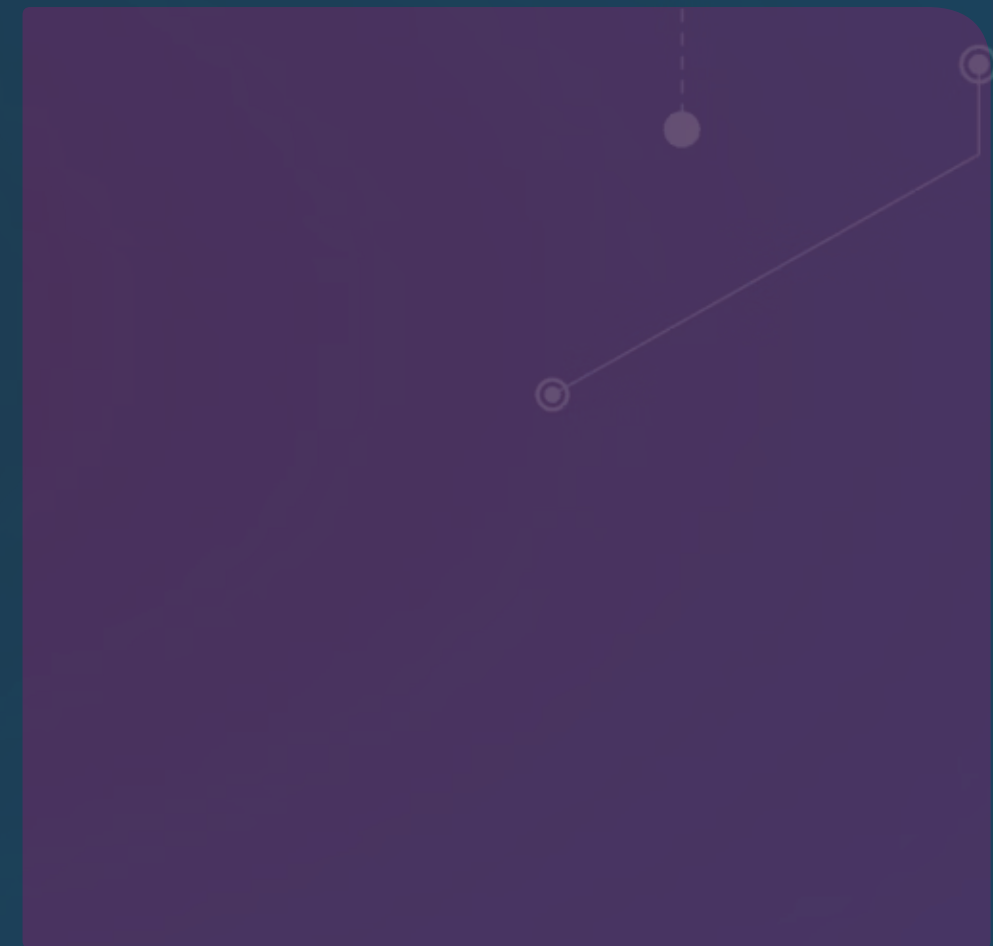
yes

7



no

12



partially



Which annotation is used to declare an entity class in JPA?

24 ✓



2 ✗

Database

1 ✗

table

Entity

Which annotation is used ignore entity field in JPA?

3 ✖

@Ignore

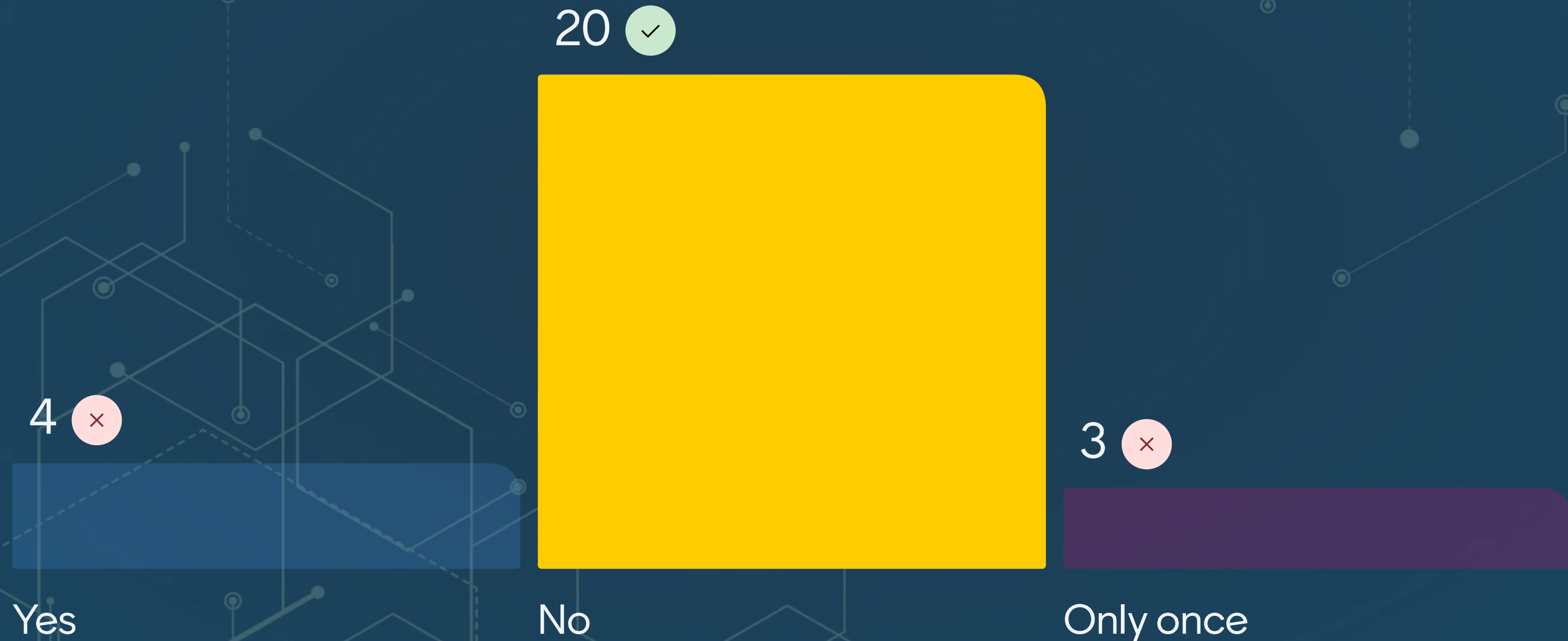
0 ✖

@Basic

23 ✔

@Transient

Can we use CriteriaQuery in HQL?



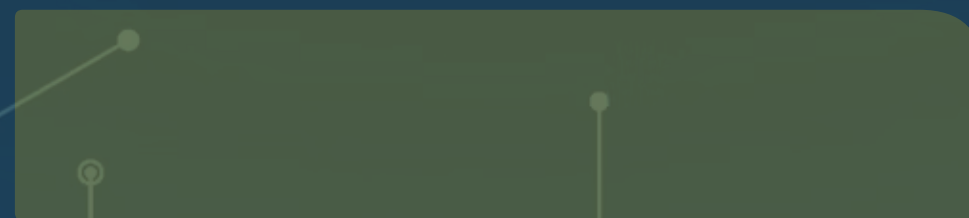
Which annotation is used store date\time related data in JPA?

5 ✗



@TimeDate

4 ✗



@Datetime

19 ✓



@Temporal

Which annotation is used store enumerations in JPA?

0



@Constant

2



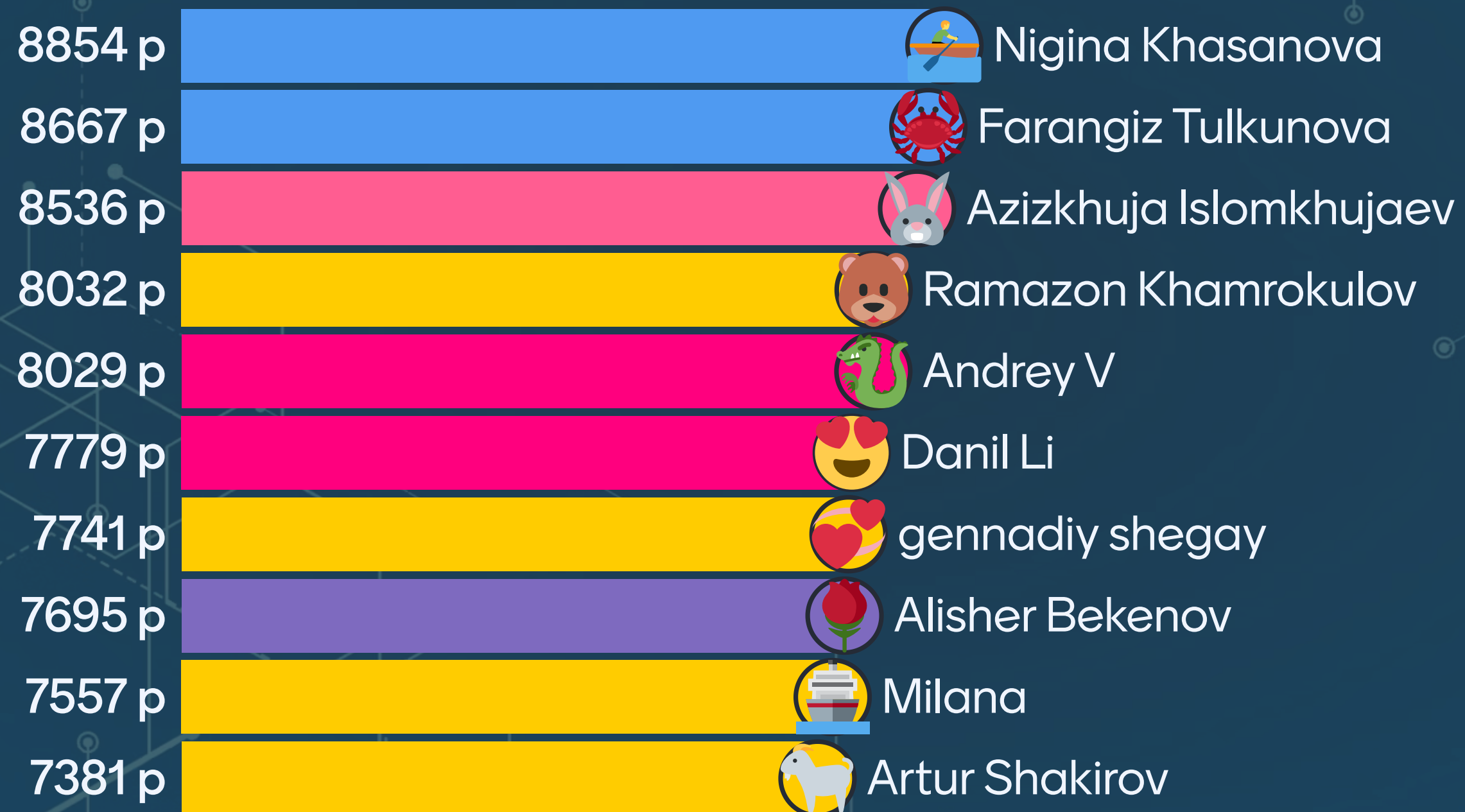
@Enum

25



@Enumerated

Quiz leaderboard



Discuss Hibernate practical task

Next time: Hibernate.

Questions?

